Sourav Ahmed

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SUMMARY

Full-Stack Software Engineer • AI/ML Enthusiast • Open Source Contributor with 3+ years at high-growth startups delivering scalable distributed systems and 0-to-1 product launches. Expert in JavaScript/TypeScript, real-time systems, and cloud infrastructure; integrating frontier LLMs into AI-assisted workflows, optimizing developer velocity by 40%.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Go (Golang), Python, Java, SQL, HTML5, CSS3 • **Frontend:** React, Next.js, Vue.js, Nuxt.js, Flutter, Tailwind CSS, Zod, shadcn/ui • **Backend:** Node.js, Express.js, .NET, GraphQL, REST API, WebSocket, Socket.IO • **Databases:** PostgreSQL, MongoDB, Redis, Firebase, BigQuery • **DevOps & Cloud:** Docker, AWS (S3, Amplify), CI/CD, Git • **AI/ML:** LLM Integration (Claude, GPT-4), TensorFlow, NLP, Computer Vision • **Tools:** Electron, OpenAPI, RBAC, Agile/Scrum

EXPERIENCE

Software Engineer II (Promoted) • Pathao Ltd. *Dec'24 - Present*

Full-stack engineer delivering scalable platforms across mapping, campaign management, and real-time systems serving 100K+ daily users while leading AI integration R&D on the Super AI team.

- **AI Integration R&D:** Leading frontier LLM research on the Super AI team — benchmarking Claude, GPT-4, Gemini, and open-source models, analyzing capability-cost tradeoffs and continuously optimizing inference costs to ship AI-assisted workflows into production using • Claude API • GPT-4 • Gemini • MCP.
- **Dashboard Foundation:** Engineered NextJs scaffold using ReactJs best practice with OpenAPI-driven type and validation, powering rapid cross-sector development with centralized auth/RBAC, and AI guardrails.
- Pathao Map (Alexander): Architected microservices-based community mapping platform with RBAC for gathering and validating POIs serving 100K+ daily users using • Go • Vue.js • Nuxt.js • Docker.
- **Loop:** Built campaign management platform handling marketing operations across all services with bulk XLSX uploads, robust file handling, and real-time analytics — improving campaign launch efficiency using • Vue.js.
- **Mission:** Developed event-driven gamification platform rewarding users for action streaks with real-time analytics dashboard for mission and goal management using • Next.js • Go • PostgreSQL • Redis • BigQuery.
- Resto Web: Engineered restaurant management SaaS with WebSocket real-time notifications, dynamic menus, and analytics dashboard using • Next.js • React • Zod • TypeScript.
- Year End Wrap: Engineered interactive year-end wrap experience shipped without app updates, solving critical cross-platform (iOS/Android) compatibility issues using • Next.js • React • TypeScript.
- **GariKoi:** Developed real-time geolocation tracking system visualizing driver locations and allocation/deallocation for operations teams using • Next.js • Go • Google Maps API.
- **Mapsense:** Built Chrome Extension and web tool to detect route anomalies by comparing actual vs suggested routes against prominent map providers using • Next.js • TypeScript • PostgreSQL.

Associate Software Engineer • ShellBeeHaken Ltd. *Aug'23 - Nov'24*

Promoted to full-time, delivered 4+ production apps across diverse business domains using React, Next.js, and Node.js.

- Mapage: Built real-time communication system for reservation/payment platform with optimistic updates and conflict resolution using • Socket.IO • Next.js • React • MongoDB.
- UUORK: Engineered cross-platform desktop workspace app (Electron + Next.js) with JWT authentication, real-time collaboration, and responsive design.
- Honnemir: Developed multimedia voting platform with • GraphQL API • AWS Amplify • S3 upload pipeline • admin moderation dashboard.
- KriyaKarak: Implemented automated payment disbursement system with reconciliation dashboard for creative marketplace using • Next.js • React • MongoDB.

Software Engineer Intern • ShellBeeHaken Ltd. *Feb'23 - Aug'23*

Shipped features and fixed critical bugs across multiple production codebases with agile processes.

PROJECTS

uBookSharing
University textbook marketplace with rental/purchase, ratings, Firebase real-time sync, custom Rive animations. • Flutter • Firebase • Rive

Obstacle Detection
Accessibility app using TensorFlow Lite for real-time object detection with audio warnings for visually impaired.
• Java • TensorFlow Lite • Computer Vision

Flash Nvim for VSCode
Published extension porting flash.nvim navigation to VS Code with 1500+ installs (also on **Open VSX** for VSCodium/Cursor), 5-star rating, intelligent search, multi-cursor. • TypeScript

Hall Management
Full-stack dormitory system with form validation, issue tracking, notice board, payment processing, student verification, RESTful API. • ReactJs • ExpressJs • .NET • MySQL

Sync Play
Collaborative video platform with synchronized playback, chat, local/cloud media support using WebSocket real-time.
• Next.js • Socket.IO • Firebase

FTP Searcher
Cross-platform FTP aggregator with search indexing, content discovery for local network sharing. • Next.js • Electron

EDUCATION

Shahjalal University of Science and Technology, Sylhet, Bangladesh *Jan'19 - Jan'24*

Institute of Information and Communication Technology (IICT)

Bachelor of Science in Software Engineering • **CGPA: 3.64/4.00** • Ranked 2nd in class (all courses first attempt)

PUBLICATIONS

IPA Transcription of Bengali Texts (arXiv 2024) — Standardized Bengali IPA phonetic transcription with novel dataset and deep learning model advancing NLP for low-resource languages. • Python • TensorFlow • NLP • Pandas

DATASETS

DUALIPA: A comprehensive Bengali IPA dataset featuring 156k+ validated sentences with phonetic transcriptions generated via deep learning models and expert verification. • **Literature (103k)** • **Newspaper (53k)**

Regional IPA: Covering phonetic variations across 7 major regions (Rangpur, Kishoreganj, Chittagong, Narsingdi, Narail, Habiganj, Tangail) with sentence and word-level transcriptions for dialectal analysis. • **Dataset**

ACHIEVEMENTS & CO-CURRICULAR

- **BdApps National Hackathon 2022 • 1st place in Regional Round 3: Sylhet, out of 100+ teams.** *• Aug'22*
- **Mobile App Game & Job Festival 2023 • 1st Runner Up in Idea Competition** *• Mar'23*
- **SJ Innovation Hackathon • Innovative Idea Prize in Medical Category** *• Oct'21*
- **Creative Executive at SUST ACM Student Chapter** *• Jul'22 - Jan'23*
- **Assistant Treasurer at Maavoi Abritti Samsad, SUST** *• Jan'20 - Jul'20*
- **Assistant Office Secretary at Manikganj Shamprity, SUST** *• Jan'22 - Aug'22*